

INTERFACE CONTROL DOCUMENT

System Communication Protocol

Version 1.0

1. SCOPE

This ICD defines the message formats and communication protocol between subsystems for the embedded platform (Version 1.0).

2. MESSAGE FORMAT

All messages use the following structure:

- msg_id : uint8 (message identifier)
- seq_num : uint8 (sequence counter 0-255)
- payload_len: uint16 (length of payload in bytes)
- payload : uint8[] (up to 64 bytes)
- checksum : uint16 (sum of all preceding fields)

3. MESSAGE IDENTIFIERS

MSG_ID_HEARTBEAT = 0x01 Periodic heartbeat
MSG_ID_STATUS = 0x02 System status report
MSG_ID_COMMAND = 0x03 Command message

4. STATUS REPORT PAYLOAD

The status report contains:

- system_id : uint8 (originating system)
- state : enum (IDLE=0, ACTIVE=1, ERROR=2)
- uptime_sec : uint16 (seconds since boot)
- temperature : int8 (degrees Celsius)

5. CHECKSUM

Simple additive checksum: sum of msg_id + seq_num + payload_len
+ all payload bytes, truncated to uint16.

6. TIMING

Heartbeat interval: 1000 ms
Status report interval: 5000 ms
Command timeout: 2000 ms